

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: movies = pd.read_csv('tmdb_5000_movies.csv')
credits = pd.read_csv('tmdb_5000_credits.csv')
```

```
In [3]: movies.head()
```

```
Out[3]:
```

	budget	genres	homepage	id	keywords	original_l
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id": ...	
1	300000000	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	
2	245000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.sonypictures.com/movies/spectre/	206647	[{"id": 470, "name": "spy"}, {"id": 818, "name...	
3	250000000	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	http://www.thedarkknightises.com/	49026	[{"id": 849, "name": "dc comics"}, {"id": 853,...	
4	260000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://movies.disney.com/john-carter	49529	[{"id": 818, "name": "based on novel"}, {"id": ...	

```
In [4]: movies.head(1)
```

```
Out[4]:
```

	budget	genres	homepage	id	keywords	original_language	original_t
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id": ...	en	Av

```
In [5]: credits.head(1)
```

```
Out[5]:
```

	movie_id	title	cast	crew
0	19995	Avatar	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...

```
In [6]: movies.shape
```

```
Out[6]: (4803, 20)
```

```
In [7]: credits.shape
```

```
Out[7]: (4803, 4)
```

```
In [8]: movies = movies.merge(credits, on='title')
```

```
In [9]: movies.shape
```

```
Out[9]: (4809, 23)
```

```
In [10]: movies.head()
```

```
Out[10]:
```

	budget	genres	homepage	id	keywords	original_l
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id": ...	
1	300000000	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	
2	245000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.sonypictures.com/movies/spectre/	206647	[{"id": 470, "name": "spy"}, {"id": 818, "name...	
3	250000000	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	http://www.thedarkknightises.com/	49026	[{"id": 849, "name": "dc comics"}, {"id": 853,...	
4	260000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://movies.disney.com/john-carter	49529	[{"id": 818, "name": "based on novel"}, {"id": ...	

5 rows × 23 columns



```
In [11]: movies.head(1)
```

```
Out[11]:
```

	budget	genres	homepage	id	keywords	original_language	original_t
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id": ...	en	Avæ

1 rows × 23 columns



```
In [12]: #genres
#id
#keywords
#title
#overview
#cast
#crew

movies = movies[['movie_id', 'title', 'overview', 'genres', 'keywords', 'cast', 'crew']]
```

```
In [13]: movies.head()
```

```
Out[13]:
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[{"id": 28, "name": "Action"}, {"id": 12, "name": "Sci-Fi"}]	[{"id": 1463, "name": "culture clash"}, {"id": 1464, "name": "space"}]	[{"cast_id": 242, "character": "Jake Sully", "credit_id": "52fe48009251416c750aca2d"}]	
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[{"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}]	[{"id": 270, "name": "ocean"}, {"id": 726, "name": "pirates"}]	[{"cast_id": 4, "character": "Captain Jack Sparrow", "credit_id": "52fe4232c3a36847f800b57d"}]	
2	206647	Spectre	A cryptic message from Bond's past sends him on a dangerous mission...	[{"id": 28, "name": "Action"}, {"id": 12, "name": "Thriller"}]	[{"id": 470, "name": "spy"}, {"id": 818, "name": "mystery"}]	[{"cast_id": 1, "character": "James Bond", "credit_id": "54805967c3a36829b5002c4d"}]	
3	49026	The Dark Knight Rises	Following the death of District Attorney Harvey Dent, Batman dedicates himself to eradicating the crime in Gotham City...	[{"id": 28, "name": "Action"}, {"id": 80, "name": "Drama"}]	[{"id": 849, "name": "dc comics"}, {"id": 853, "name": "superhero"}]	[{"cast_id": 2, "character": "Bruce Wayne / Batman", "credit_id": "52fe4781c3a36847f81398cd"}]	
4	49529	John Carter	John Carter is a war-weary, former military captain who has been living in a small town in the American West...	[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}]	[{"id": 818, "name": "based on novel"}, {"id": 819, "name": "science fiction"}]	[{"cast_id": 5, "character": "John Carter", "credit_id": "52fe479ac3a36847f813eae2"}]	

```
In [14]: movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 4809 entries, 0 to 4808
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   movie_id    4809 non-null   int64
1   title       4809 non-null   object
2   overview    4806 non-null   object
3   genres      4809 non-null   object
4   keywords    4809 non-null   object
5   cast        4809 non-null   object
6   crew        4809 non-null   object
dtypes: int64(1), object(6)
memory usage: 300.6+ KB
```

```
In [15]: movies.isnull().sum()
```

```
Out[15]: movie_id    0
         title      0
         overview    3
         genres     0
         keywords    0
         cast       0
         crew       0
         dtype: int64
```

```
In [16]: movies.dropna(inplace=True)
```

```
In [17]: movies.duplicated().sum()
```

```
Out[17]: 0
```

```
In [18]: movies.iloc[0].genres
```

```
Out[18]: '[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}, {"id": 878, "name": "Science Fiction"}]'
```

```
In [19]: #'[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}, {"id": 878, "name": "Science Fiction"}]'
         #[action,adventure,fantasy,scifi]
```

```
In [20]: import ast
         def convert(obj):
             L=[]
             for i in ast.literal_eval(obj):
                 L.append(i['name'])
             return L
```

```
In [21]: movies['genres']= movies['genres'].apply(convert)
```

```
In [22]: movies['genres']
```

```
Out[22]: 0      [Action, Adventure, Fantasy, Science Fiction]
1      [Adventure, Fantasy, Action]
2      [Action, Adventure, Crime]
3      [Action, Crime, Drama, Thriller]
4      [Action, Adventure, Science Fiction]
...
4804      [Action, Crime, Thriller]
4805      [Comedy, Romance]
4806      [Comedy, Drama, Romance, TV Movie]
4807      []
4808      [Documentary]
Name: genres, Length: 4806, dtype: object
```

```
In [23]: movies.head()
```

```
Out[23]:
```

	movie_id	title	overview	genres	keywords	cast	cre
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[{"id": 1463, "name": "culture clash"}, {"id": ...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de.
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit_id": "52fe4232c3a36847f800b579", "de.
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[{"id": 470, "name": "spy"}, {"id": 818, "name...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit_id": "54805967c3a36829b5002c41", "de.
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[{"id": 849, "name": "dc comics"}, {"id": 853, ...	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"credit_id": "52fe4781c3a36847f81398c3", "de.
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[{"id": 818, "name": "based on novel"}, {"id": ...	[{"cast_id": 5, "character": "John Carter", "c...	[{"credit_id": "52fe479ac3a36847f81398c3", "de.

```
In [24]: movies['keywords']= movies['keywords'].apply(convert)
```

```
In [25]: movies.head()
```

0	1995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit": "52fe48009251416c750aca", ...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit": "52fe4232c3a36847f800b5", ...
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit": "54805967c3a36829b5002c", ...

```
In [26]: movies['cast']
```

```
Out[26]: 0      [{"cast_id": 242, "character": "Jake Sully", "...
1      [{"cast_id": 4, "character": "Captain Jack Spa...
2      [{"cast_id": 1, "character": "James Bond", "cr...
3      [{"cast_id": 2, "character": "Bruce Wayne / Ba...
4      [{"cast_id": 5, "character": "John Carter", "c...
...
4804   [{"cast_id": 1, "character": "El Mariachi", "c...
4805   [{"cast_id": 1, "character": "Buzzy", "credit_...
4806   [{"cast_id": 8, "character": "Oliver O\u00d919To...
4807   [{"cast_id": 3, "character": "Sam", "credit_id...
4808   [{"cast_id": 3, "character": "Herself", "credi...
Name: cast, Length: 4806, dtype: object
```

```
In [27]: import ast
def convert3(obj):
    L=[]
    counter = 0
    for i in ast.literal_eval(obj):
        if counter!=3:
            L.append(i['name'])
            counter = counter+1
        else:
            break
    return L
```



```
In [28]: movies['cast'] =movies['cast'].apply(convert3)
```

```
In [29]: movies.head()
```

Out[29]:

	movie_id	title	overview	genres	keywords	cast	cr
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[{"credit_i "52fe48009251416c750aca2: "de
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[{"credit_i "52fe4232c3a36847f800b57: "de
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[{"credit_i "54805967c3a36829b5002c4 "de
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret i...	[Christian Bale, Michael Caine, Gary Oldman]	[{"credit_i "52fe4781c3a36847f81398c "de
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[based on novel, mars, medallion, space travel...	[Taylor Kitsch, Lynn Collins, Samantha Morton]	[{"credit_i "52fe479ac3a36847f813eaa "de




```
In [30]: movies['crew'][0]
```

```
Out[30]: '[{"credit_id": "52fe48009251416c750aca23", "department": "Editing", "gender": 0, "id": 1721, "job": "Editor", "name": "Stephen E. Rivkin"}, {"credit_id": "539c47ecc3a36810e3001f87", "department": "Art", "gender": 2, "id": 496, "job": "Production Design", "name": "Rick Carter"}, {"credit_id": "54491c89c3a3680fb4001cf7", "department": "Sound", "gender": 0, "id": 900, "job": "Sound Designer", "name": "Christopher Boyes"}, {"credit_id": "54491cb70e0a267480001bd0", "department": "Sound", "gender": 0, "id": 900, "job": "Supervising Sound Editor", "name": "Christopher Boyes"}, {"credit_id": "539c4a4cc3a36810c9002101", "department": "Production", "gender": 1, "id": 1262, "job": "Casting", "name": "Mali Finn"}, {"credit_id": "5544ee3b925141499f0008fc", "department": "Sound", "gender": 2, "id": 1729, "job": "Original Music Composer", "name": "James Horner"}, {"credit_id": "52fe48009251416c750ac9c3", "department": "Directing", "gender": 2, "id": 2710, "job": "Director", "name": "James Cameron"}, {"credit_id": "52fe48009251416c750ac9d9", "department": "Writing", "gender": 2, "id": 2710, "job": "Writer", "name": "James Cameron"}, {"credit_id": "52fe48009251416c750aca17", "department": "Editing", "gender": 2, "id": 2710, "job": "Editor", "name": "James Cameron"}, {"credit_id": "52fe48009251416c750aca29", "department": "Production", "gender": 2, "id": 2710, "job": "Producer", "name": "James Cameron"}]
```

```
In [31]: import ast
def fetch_director(obj):
    L=[]
    for i in ast.literal_eval(obj):
        if i['job']=='Director':
            L.append(i['name'])
            break
    return L
```

```
In [32]: movies['crew'] = movies['crew'].apply(fetch_director)
```

```
In [33]: movies.head()
```

```
Out[33]:
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[James Cameron]
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[Gore Verbinski]
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[Sam Mendes]
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret i...	[Christian Bale, Michael Caine, Gary Oldman]	[Christopher Nolan]
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[based on novel, mars, medallion, space travel...	[Taylor Kitsch, Lynn Collins, Samantha Morton]	[Andrew Stanton]

```
In [34]: movies['overview'][0]
```

```
Out[34]: 'In the 22nd century, a paraplegic Marine is dispatched to the moon Pandora o
n a unique mission, but becomes torn between following orders and protecting
an alien civilization.'
```

```
In [35]: movies['overview'] = movies['overview'].apply(lambda x:x.split())
```

In [36]: `movies.head()`

Out[36]:

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	[In, the, 22nd, century,, a, paraplegic, Marin...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[James Cameron]
1	285	Pirates of the Caribbean: At World's End	[Captain, Barbossa,, long, believed, to, be, d...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[Gore Verbinski]
2	206647	Spectre	[A, cryptic, message, from, Bond's, past, send...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[Sam Mendes]
3	49026	The Dark Knight Rises	[Following, the, death, of, District, Attorney...	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret i...	[Christian Bale, Michael Caine, Gary Oldman]	[Christopher Nolan]
4	49529	John Carter	[John, Carter, is, a, war-weary,, former, mili...	[Action, Adventure, Science Fiction]	[based on novel, mars, medallion, space travel...	[Taylor Kitsch, Lynn Collins, Samantha Morton]	[Andrew Stanton]

In [37]: `movies['genres'] = movies['genres'].apply(lambda x:[i.replace(" ", "") for i in x])`
`movies['keywords'] = movies['keywords'].apply(lambda x:[i.replace(" ", "") for i in x])`
`movies['cast'] = movies['cast'].apply(lambda x:[i.replace(" ", "") for i in x])`
`movies['crew'] = movies['crew'].apply(lambda x:[i.replace(" ", "") for i in x])`

```
In [38]: movies.head()
```

```
Out[38]:
```

	movie_id	title	overview	genres	keywords	cast	c
0	19995	Avatar	[In, the, 22nd, century,, a, paraplegic, Marin...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesCame
1	285	Pirates of the Caribbean: At World's End	[Captain, Barbossa,, long, believed, to, be, d...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreVerbi
2	206647	Spectre	[A, cryptic, message, from, Bond's, past, send...	[Action, Adventure, Crime]	[spy, basedonnovel, secretagent, sequel, mi6, ...	[DanielCraig, ChristophWaltz, LéaSeydoux]	[SamMen
3	49026	The Dark Knight Rises	[Following, the, death, of, District, Attorney...	[Action, Crime, Drama, Thriller]	[dccomics, crimefighter, terrorist, secretiden...	[ChristianBale, MichaelCaine, GaryOldman]	[ChristopherNc
4	49529	John Carter	[John, Carter, is, a, war-weary,, former, mili...	[Action, Adventure, ScienceFiction]	[basedonnovel, mars, medallion, spacetravel, p...	[TaylorKitsch, LynnCollins, SamanthaMorton]	[AndrewStar



```
In [39]: movies['tags'] = movies['overview']+movies['genres']+movies['keywords']+movies['cast']
```

```
In [40]: movies.head()
```

```
Out[40]:
```

	movie_id	title	overview	genres	keywords	cast	c
0	19995	Avatar	[In, the, 22nd, century,, a, paraplegic, Marin...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesCame
1	285	Pirates of the Caribbean: At World's End	[Captain, Barbossa,, long, believed, to, be, d...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreVerbi
2	206647	Spectre	[A, cryptic, message, from, Bond's, past, send...	[Action, Adventure, Crime]	[spy, basedonnovel, secretagent, sequel, mi6, ...	[DanielCraig, ChristophWaltz, LéaSeydoux]	[SamMen
3	49026	The Dark Knight Rises	[Following, the, death, of, District, Attorney...	[Action, Crime, Drama, Thriller]	[dccomics, crimefighter, terrorist, secretiden...	[ChristianBale, MichaelCaine, GaryOldman]	[ChristopherNc
4	49529	John Carter	[John, Carter, is, a, war-weary,, former, mili...	[Action, Adventure, ScienceFiction]	[basedonnovel, mars, medallion, spacetravel, p...	[TaylorKitsch, LynnCollins, SamanthaMorton]	[AndrewStar

```
In [41]: movies['tags'][0]
```

```
Out[41]: ['In',
          'the',
          '22nd',
          'century,',
          'a',
          'paraplegic',
          'Marine',
          'is',
          'dispatched',
          'to',
          'the',
          'moon',
          'Pandora',
          'on',
          'a',
          'unique',
          'mission,',
          'but',
          'becomes',
          'torn',
          'between',
          'following',
          'orders',
          'and',
          'protecting',
          'an',
          'alien',
          'civilization.',
          'Action',
          'Adventure',
          'Fantasy',
          'ScienceFiction',
          'cultureclash',
          'future',
          'spacewar',
          'spacecolony',
          'society',
          'spacetravel',
          'futuristic',
          'romance',
          'space',
          'alien',
          'tribe',
          'alienplanet',
          'cgi',
          'marine',
          'soldier',
          'battle',
          'loveaffair',
          'antiwar',
          'powerrelations',
          'mindandsoul',
          '3d',
          'SamWorthington',
          'ZoeSaldana',
```



```
'SigourneyWeaver',  
'JamesCameron']
```

```
In [42]: new_df = movies[['movie_id', 'title', 'tags']]
```

```
In [43]: new_df
```

```
Out[43]:
```

	movie_id		title	tags
0	19995		Avatar	[In, the, 22nd, century,, a, paraplegic, Marin...
1	285	Pirates of the Caribbean: At World's End		[Captain, Barbossa,, long, believed, to, be, d...
2	206647		Spectre	[A, cryptic, message, from, Bond's, past, send...
3	49026	The Dark Knight Rises		[Following, the, death, of, District, Attorney...
4	49529		John Carter	[John, Carter, is, a, war-weary,, former, mili...
...	...		...	...
4804	9367		El Mariachi	[El, Mariachi, just, wants, to, play, his, gui...
4805	72766		Newlyweds	[A, newlywed, couple's, honeymoon, is, upended...
4806	231617	Signed, Sealed, Delivered		["Signed,, Sealed,, Delivered", introduces, a,...
4807	126186		Shanghai Calling	[When, ambitious, New, York, attorney, Sam, is...
4808	25975		My Date with Drew	[Ever, since, the, second, grade, when, he, fi...

4806 rows × 3 columns

```
In [44]: new_df['tags'] = new_df['tags'].apply(lambda x: " ".join(x))
```

C:\Users\lenovo\AppData\Local\Temp\ipykernel_10512\1824047427.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
new_df['tags'] = new_df['tags'].apply(lambda x: " ".join(x))
```

```
In [45]: new_df.head()
```

```
Out[45]:
```

	movie_id		title	tags
0	19995		Avatar	In the 22nd century, a paraplegic Marine is di...
1	285	Pirates of the Caribbean: At World's End		Captain Barbossa, long believed to be dead, ha...
2	206647		Spectre	A cryptic message from Bond's past sends him o...
3	49026	The Dark Knight Rises		Following the death of District Attorney Harve...
4	49529		John Carter	John Carter is a war-weary, former military ca...

```
In [46]: new_df['tags'] = new_df['tags'].apply(lambda x:x.lower())
```

C:\Users\lenovo\AppData\Local\Temp\ipykernel_10512\3214958533.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
new_df['tags'] = new_df['tags'].apply(lambda x:x.lower())
```

```
In [47]: new_df
```

```
Out[47]:
```

	movie_id		title	tags
0	19995		Avatar	in the 22nd century, a paraplegic marine is di...
1	285	Pirates of the Caribbean: At World's End		captain barbossa, long believed to be dead, ha...
2	206647		Spectre	a cryptic message from bond's past sends him o...
3	49026	The Dark Knight Rises		following the death of district attorney harve...
4	49529		John Carter	john carter is a war-weary, former military ca...
...	...		...	...
4804	9367		El Mariachi	el mariachi just wants to play his guitar and ...
4805	72766		Newlyweds	a newlywed couple's honeymoon is upended by th...
4806	231617	Signed, Sealed, Delivered		"signed, sealed, delivered" introduces a dedic...
4807	126186		Shanghai Calling	when ambitious new york attorney sam is sent t...
4808	25975	My Date with Drew		ever since the second grade when he first saw ...

4806 rows × 3 columns

```
In [48]: new_df['tags'][0]
```

```
Out[48]: 'in the 22nd century, a paraplegic marine is dispatched to the moon pandora on a unique mission, but becomes torn between following orders and protecting an alien civilization. action adventure fantasy sciencefiction cultureclash future spacewar spacecolony society spacetravel futuristic romance space alien tribe alienplanet cgi marine soldier battle loveaffair antiwar powerrelations mindandsoul 3d samworthington zoesaldana sigourneyweaver jamescameron'
```

```
In [49]: new_df['tags'][1]
```

```
Out[49]: "captain barbossa, long believed to be dead, has come back to life and is headed to the edge of the earth with will turner and elizabeth swann. but nothing is quite as it seems. adventure fantasy action ocean drugabuse exoticisland eastindiatradingcompany loveofone'slife traitor shipwreck strongwoman ship alliance calypso afterlife fighter pirate swashbuckler aftercreditsstinger john nydepp orlandobloom keiraknightley goreverbinski"
```

```
In [50]: from sklearn.feature_extraction.text import CountVectorizer  
cv = CountVectorizer(max_features = 5000, stop_words= 'english')
```

```
In [51]: vectors = cv.fit_transform(new_df['tags']).toarray()
```

```
In [52]: vectors[0]
```

```
Out[52]: array([0, 0, 0, ..., 0, 0, 0], dtype=int64)
```

```
In [53]: cv.get_feature_names()
```

```
C:\ProgramData\Anaconda3\lib\site-packages\sklearn\utils\deprecation.py:87:  
FutureWarning: Function get_feature_names is deprecated; get_feature_names  
is deprecated in 1.0 and will be removed in 1.2. Please use get_feature_names_out instead.
```

```
warnings.warn(msg, category=FutureWarning)
```

```
Out[53]: ['000',  
          '007',  
          '10',  
          '100',  
          '11',  
          '12',  
          '13',  
          '14',  
          '15',  
          '16',  
          '17',  
          '18',  
          '18th',  
          '19']
```

```
In [54]: import nltk
```

```
In [55]: from nltk.stem.porter import PorterStemmer  
ps = PorterStemmer()
```

```
In [56]: ps.stem('loving')
```

```
Out[56]: 'love'
```

```
In [57]: def stem(text):
        y = []

        for i in text.split():
            y.append(ps.stem(i))

        return " ".join(y)
```

```
In [58]: new_df['tags'] = new_df['tags'].apply(stem)
```

C:\Users\lenovo\AppData\Local\Temp\ipykernel_10512\3213734980.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
new_df['tags'] = new_df['tags'].apply(stem)
```

```
In [59]: new_df
```

```
Out[59]:
```

	movie_id		title	tags
0	19995		Avatar	in the 22nd century, a parapleg marin is dispa...
1	285	Pirates of the Caribbean: At World's End		captain barbossa, long believ to be dead, ha c...
2	206647		Spectre	a cryptic messag from bond' past send him on a...
3	49026	The Dark Knight Rises		follow the death of district attorney harvey d...
4	49529		John Carter	john carter is a war-weary, former militari ca...
...	...		...	...
4804	9367		El Mariachi	el mariachi just want to play hi guitar and ca...
4805	72766		Newlyweds	a newlyw couple' honeymoon is upend by the arr...
4806	231617	Signed, Sealed, Delivered		"signed, sealed, delivered" introduc a dedic q...
4807	126186		Shanghai Calling	when ambiti new york attorney sam is sent to s...
4808	25975		My Date with Drew	ever sinc the second grade when he first saw h...

4806 rows × 3 columns

```
In [60]: from sklearn.metrics.pairwise import cosine_similarity
```

```
In [61]: similarity = cosine_similarity(vectors)
```

```
In [62]: similarity[2]
```

```
Out[62]: array([0.05976143, 0.0625      , 1.          , ..., 0.02635231, 0.          ,
                0.          ])
```

```
In [63]: similarity[0]
```

```
Out[63]: array([1.          , 0.08964215, 0.05976143, ..., 0.02519763, 0.02817181,
                0.          ])
```

```
In [64]: sorted(similarity[0], reverse = True)
```

```
Out[64]: [1.0,
          0.26089696604360174,
          0.2581988897471611,
          0.25302403842552984,
          0.25110592822973776,
          0.24944382578492943,
          0.24397501823713333,
          0.243599382882345,
          0.23904572186687872,
          0.2344036154692477,
          0.2267786838055363,
          0.2238868314198225,
          0.22230800575069135,
          0.22190115272469205,
          0.22131333406899523,
          0.2193633988428327,
          0.2173253797873328,
          0.21653278478430668,
          0.2164218276749025,
          0.2163340010106011]
```

```
In [65]: list(enumerate(similarity[0]))
```

```
Out[65]: [(0, 1.0),
          (1, 0.08964214570007951),
          (2, 0.059761430466719674),
          (3, 0.03984095364447979),
          (4, 0.18257418583505536),
          (5, 0.1183452670827877),
          (6, 0.02258769757263128),
          (7, 0.1763834207376394),
          (8, 0.061721339984836754),
          (9, 0.07644707871564382),
          (10, 0.11769797726729994),
          (11, 0.081199794294115),
          (12, 0.09583148474999097),
          (13, 0.04733810683311508),
          (14, 0.11428571428571427),
          (15, 0.05096471914376255),
          (16, 0.08226127456606226),
          (17, 0.15026857675938213),
          (18, 0.1129384878631564),
          (19, 0.02730715600430011)]
```



```
In [66]: def recommend(movie):
         movie_index = new_df[new_df['title']== movie].index[0]
         distances= similarity[movie_index]
         movies_list= sorted(list(enumerate(distances)), reverse = True, key = lambda
                                x:x[1])

         for i in movies_list:
             print(new_df.iloc[i[0]].title)
```

```
In [75]: recommend('Avatar')
```

Titan A.E.
Small Soldiers
Independence Day
Ender's Game
Aliens vs Predator: Requiem

```
In [68]: new_df.iloc[539].title
```

```
Out[68]: 'Titan A.E.'
```

```
In [69]: sorted(list(enumerate(similarity[0])), reverse = True, key=lambda x:x[1])[1:6]
```

```
Out[69]: [(539, 0.26089696604360174),
          (1194, 0.2581988897471611),
          (507, 0.25302403842552984),
          (260, 0.25110592822973776),
          (1216, 0.24944382578492943)]
```

```
In [70]: new_df[new_df['title']=="Batman Begins"].index[0]
```

```
Out[70]: 119
```

```
In [71]: import pickle
```

```
In [72]: pickle.dump(new_df.to_dict(),open('movie_dict.pkl', 'wb'))
```

In [73]: new_df

Out[73]:

	movie_id		title	tags
0	19995		Avatar	in the 22nd century, a parapleg marin is dispa...
1	285	Pirates of the Caribbean: At World's End		captain barbossa, long believ to be dead, ha c...
2	206647		Spectre	a cryptic messag from bond' past send him on a...
3	49026	The Dark Knight Rises		follow the death of district attorney harvey d...
4	49529		John Carter	john carter is a war-weary, former militari ca...
...	...		...	...
4804	9367		El Mariachi	el mariachi just want to play hi guitar and ca...
4805	72766		Newlyweds	a newlyw couple' honeymoon is upend by the arr...
4806	231617	Signed, Sealed, Delivered		"signed, sealed, delivered" introduc a dedic q...
4807	126186		Shanghai Calling	when ambiti new york attorney sam is sent to s...
4808	25975		My Date with Drew	ever sinc the second grade when he first saw h...

4806 rows × 3 columns

In [74]: pickle.dump(similarity,open('similarity.pkl','wb'))

In []:

In []: